

UNIVERSITY OF BRISTOL

Faculty of Science

**MOCK EXAMINATION — Paper 6 (Easy) —
SOLUTIONS AND MARK SCHEME****Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012

Total Marks: 100

Marking Notes:

- Mark allocations are shown in **blue** at each step.
 - Final answers are highlighted in green boxes.
 - “(*From formula sheet*)” indicates the equation is on the provided formula sheet.
 - “(*Recalled — not on formula sheet*)” indicates the equation must be known from memory.
 - Award partial marks for correct method even if the final answer is wrong.
 - Accept equivalent correct forms of answers.
-

Section A: Oscillations and Waves — Answers

1. **B** — Simple harmonic motion is defined by a restoring force that is directly proportional to the displacement from equilibrium and directed toward the equilibrium position: $F = -kx$. This linear restoring force is the defining property of SHM.
2. **C** — The period of a mass-spring system is $T = 2\pi\sqrt{m/k}$ (*From formula sheet*). With $m = 0.5$ kg and $k = 200$ N/m: $T = 2\pi\sqrt{0.5/200} = 2\pi\sqrt{0.0025} = 2\pi \times 0.05 = 0.314$ s.
3. **B** — The maximum speed in SHM is $v_{\max} = A\omega_0$ (*Recalled — not on formula sheet*). With $A = 0.10$ m and $\omega_0 = 5$ rad/s: $v_{\max} = 0.10 \times 5 = 0.50$ m/s.
4. **B** — The quality factor is defined as $Q = \omega_0/\gamma$ (*From formula sheet*), the ratio of the natural angular frequency to the damping constant.
5. **B** — The wave speed is $v = f\lambda$ (*Recalled — not on formula sheet*). With $f = 500$ Hz and $\lambda = 0.68$ m: $v = 500 \times 0.68 = 340$ m/s.
6. **C** — In the n th harmonic of a string fixed at both ends, there are n antinodes (and $n + 1$ nodes). For the 3rd harmonic, there are 3 antinodes.
7. **C** — The decibel formula is $\beta = 10\log_{10}(I/I_0)$ (*From formula sheet*), where $I_0 = 10^{-12}$ W/m² is the reference intensity. At $\beta = 0$ dB: $0 = 10\log_{10}(I/I_0)$, so $I/I_0 = 1$, giving $I = I_0 = 10^{-12}$ W/m².
8. **C** — When a source moves toward an observer, the effective wavelength is compressed (shorter), resulting in a higher observed frequency. From the Doppler formula $f' = f v/(v - v_s)$ (*From formula sheet*), with $v_s > 0$ (approaching), the denominator is less than v , so $f' > f$.
9. **B** — The beat frequency is $f_{\text{beat}} = |f_1 - f_2|$ (*Recalled — not on formula sheet*). With $f_1 = 256$ Hz and $f_2 = 260$ Hz: $f_{\text{beat}} = |256 - 260| = 4$ Hz.
10. **B** — The period of a simple pendulum is $T = 2\pi\sqrt{L/g}$ (*From formula sheet*). It depends only on the length L of the string and the acceleration due to gravity g , not on the mass of the bob or the amplitude (for small oscillations).

Section B: Quantum Physics — Answers

1. **C** — The symbol $|\psi\rangle$ is a ket, which represents a quantum state (state vector) in a Hilbert space. It is not a number, a probability, or an operator.
2. **D** — The inner product $\langle\phi|\psi\rangle$ is a complex scalar (a complex number). It is the “overlap” between the two states. (*Recalled — not on formula sheet*)
3. **B** — By the Born rule, $|\langle\phi|\psi\rangle|^2$ gives the probability of finding the system in state $|\phi\rangle$ given it is in state $|\psi\rangle$ (or equivalently, the transition/overlap probability). (*Recalled — not on formula sheet*)
4. **C** — In the eigenvalue equation $\hat{A}|a\rangle = a|a\rangle$, the scalar a is called the eigenvalue (and $|a\rangle$ is the eigenvector or eigenstate). (*Recalled — not on formula sheet*)
5. **B** — For a spin- $\frac{1}{2}$ particle, the eigenvalues of S_z are $+\hbar/2$ (spin up) and $-\hbar/2$ (spin down). These follow from $S_z = \frac{\hbar}{2}\sigma_z$ (*From formula sheet*).
6. **C** — The normalisation condition requires $\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1$ (*Recalled — not on formula sheet*). Note the modulus squared, not the square of the coefficients (since α and β may be complex).
7. **C** — An operator is Hermitian (self-adjoint) if $\hat{A}^\dagger = \hat{A}$. This ensures its eigenvalues are real, which is required for physical observables. (*Recalled — not on formula sheet*)
8. **C** — For spin quantum number s , the number of distinct spin states is $2s + 1$. For $s = 1$: $2(1) + 1 = 3$ states, corresponding to $m_s = -1, 0, +1$.
9. **C** — The proton (uud) has charge $+\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = +\frac{4}{3}e - \frac{1}{3}e = +1e = +e$. (*Recalled — not on formula sheet*)
10. **B** — An electron is a fermion (it has spin- $\frac{1}{2}$) and a lepton (it is a fundamental particle in the lepton family, not composed of quarks). (*Recalled — not on formula sheet*)

Section C: Long Answer Solutions

Question 1.

[15 marks]

SUVAT, Forces, and Energy — Direct Application

- (a) (i) At maximum height, $v = 0$. Using $v = u + at$ with $u = 20$ m/s, $a = -g = -9.81$ m/s²: (*From formula sheet*)

$$0 = 20 - 9.81t$$

$$t = \frac{20}{9.81} = 2.04 \text{ s}$$

[1]

- (ii) Using $v^2 = u^2 + 2as$ with $v = 0$: (*From formula sheet*)

$$0 = 20^2 + 2(-9.81)h$$

$$h = \frac{400}{2 \times 9.81} = \frac{400}{19.62}$$

$$h = 20.4 \text{ m}$$

[1]

- (iii) By symmetry, the total time of flight is twice the time to reach maximum height:

$$t_{\text{total}} = 2 \times 2.04 = 4.08 \text{ s}$$

[1]

(Alternatively, use $s = ut + \frac{1}{2}at^2$ with $s = 0$: $0 = 20t - \frac{1}{2}(9.81)t^2$, giving $t(20 - 4.905t) = 0$, so $t = 0$ or $t = 20/4.905 = 4.08$ s.)

- (b) (i) Forces acting on the 10 kg box:

[1]

- Weight $W = mg = 10 \times 9.81 = 98.1$ N downward.
- Normal force $N = 98.1$ N upward (balances weight on horizontal surface).
- Applied force $F = 60$ N horizontal, in the direction of motion.
- Kinetic friction f_k opposing the motion (horizontal, opposite to applied force).

- (ii) The kinetic friction force is: (*Recalled — not on formula sheet*)

$$f_k = \mu_k N = \mu_k mg = 0.4 \times 10 \times 9.81$$

$$f_k = 39.2 \text{ N}$$

[1]

- (iii) Applying Newton's second law ($F_{\text{net}} = ma$) horizontally: (*Recalled — not on formula sheet*)

$$F - f_k = ma$$

$$60 - 39.2 = 10a$$

$$a = \frac{20.8}{10}$$

$$a = 2.08 \text{ m/s}^2$$

[1]

- (c) (i) Using conservation of energy for the ball falling from height $h = 5 \text{ m}$: (*Recalled — not on formula sheet*)

$$mgh = \frac{1}{2}mv^2$$

$$v = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 5} = \sqrt{98.1}$$

[1]

$$v = 9.90 \text{ m/s}$$

[1]

- (ii) The ball slides onto the frictionless horizontal surface at speed $v = 9.90 \text{ m/s}$ and compresses the spring. At maximum compression x , all kinetic energy is stored as elastic potential energy: (*Recalled — not on formula sheet*)

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$

[1]

$$x = v\sqrt{\frac{m}{k}} = 9.90 \times \sqrt{\frac{2}{500}} = 9.90 \times \sqrt{0.004} = 9.90 \times 0.06325$$

$$x = 0.626 \text{ m} \approx 0.63 \text{ m}$$

[1]

- (d) (i) Perfectly inelastic collision: the objects stick together. Conservation of momentum: (*Recalled — not on formula sheet*)

$$m_1v_1 + m_2v_2 = (m_1 + m_2)v_f$$

[1]

$$3 \times 4 + 1 \times 0 = (3 + 1)v_f$$

$$12 = 4v_f$$

$$v_f = 3 \text{ m/s (to the right)}$$

[1]

- (ii) Kinetic energy before: (*Recalled — not on formula sheet*)

$$\text{KE}_{\text{before}} = \frac{1}{2}m_1v_1^2 = \frac{1}{2}(3)(4^2) = 24 \text{ J}$$

Kinetic energy after:

$$\text{KE}_{\text{after}} = \frac{1}{2}(m_1 + m_2)v_f^2 = \frac{1}{2}(4)(3^2) = 18 \text{ J}$$

$$\text{Kinetic energy lost} = 24 - 18 = 6 \text{ J}$$

[1]

(iii) For an elastic collision, the formula sheet gives: (*From formula sheet*)

$$v'_1 = \frac{m_1 - m_2}{m_1 + m_2} v_1 + \frac{2m_2}{m_1 + m_2} v_2$$

By analogy (swapping indices 1 and 2):

$$v'_2 = \frac{2m_1}{m_1 + m_2} v_1 + \frac{m_2 - m_1}{m_1 + m_2} v_2 \quad [1]$$

Since $v_2 = 0$:

$$v'_2 = \frac{2m_1}{m_1 + m_2} v_1 = \frac{2 \times 3}{3 + 1} \times 4 = \frac{6}{4} \times 4 = 6 \text{ m/s}$$

$$v'_2 = 6 \text{ m/s (to the right)}$$

[1]

Verification: $v'_1 = \frac{3-1}{3+1} \times 4 + 0 = \frac{2}{4} \times 4 = 2 \text{ m/s}$. Momentum: $3 \times 2 + 1 \times 6 = 12 \checkmark$.
KE: $\frac{1}{2}(3)(4) + \frac{1}{2}(1)(36) = 6 + 18 = 24 \text{ J } \checkmark$.

Question 2.

[15 marks]

Heat, Ideal Gas, and Basic Thermodynamics

- (a) Energy to heat water from 20 °C to 100 °C: (
- Recalled — not on formula sheet*
-)

$$Q_{\text{heat}} = mc \Delta T = 0.5 \times 4200 \times (100 - 20) = 0.5 \times 4200 \times 80 \quad [1]$$

$$Q_{\text{heat}} = 168,000 \text{ J}$$

Energy to vaporise water at 100 °C: (*Recalled — not on formula sheet*)

$$Q_{\text{vap}} = mL_{\text{vap}} = 0.5 \times 2.26 \times 10^6 = 1,130,000 \text{ J} \quad [1]$$

Total energy:

$$Q_{\text{total}} = Q_{\text{heat}} + Q_{\text{vap}} = 168,000 + 1,130,000 = 1,298,000 \text{ J} \approx 1.30 \text{ MJ}$$

[1]

- (b) (i) Using the ideal gas law
- $PV = nRT$
- : (
- From formula sheet*
-)

$$V = \frac{nRT}{P} = \frac{1 \times 8.314 \times 300}{1.0 \times 10^5} = \frac{2494}{1.0 \times 10^5}$$

$$V = 0.02494 \text{ m}^3 \approx 0.0249 \text{ m}^3$$

[1]

- (ii) At constant pressure,
- $V \propto T$
- (from
- $PV = nRT$
-):

$$V_2 = V_1 \times \frac{T_2}{T_1} = 0.02494 \times \frac{600}{300} = 0.02494 \times 2$$

$$V_2 = 0.04989 \text{ m}^3 \approx 0.0499 \text{ m}^3$$

[1]

- (iii) Work done by the gas in an isobaric process: (
- Recalled — not on formula sheet*
-)

$$W = P \Delta V = 1.0 \times 10^5 \times (0.04989 - 0.02494) = 1.0 \times 10^5 \times 0.02494$$

$$W = 2494 \text{ J} \approx 2490 \text{ J}$$

[1]

- (iv) Heat absorbed at constant pressure: (
- Recalled — not on formula sheet*
-)

$$Q = nC_P \Delta T$$

For monatomic ideal gas: $C_V = \frac{3}{2}R$ (*Recalled — not on formula sheet*) and $C_P = C_V + R = \frac{5}{2}R = \frac{5}{2} \times 8.314 = 20.79 \text{ J/(mol K)}$ (*From formula sheet*)

$$Q = 1 \times 20.79 \times (600 - 300) = 1 \times 20.79 \times 300$$

$$Q = 6236 \text{ J} \approx 6240 \text{ J}$$

[1]

Check via first law: $\Delta U = nC_V \Delta T = 1 \times \frac{3}{2} \times 8.314 \times 300 = 3741 \text{ J}$. $Q - W = 6236 - 2494 = 3742 \text{ J} \approx \Delta U \checkmark$

(c) (i) For an adiabatic process: $TV^{\gamma-1} = \text{const}$ (*From formula sheet*)

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

With $\gamma - 1 = 2/3$, $T_1 = 600 \text{ K}$, $V_1 = 0.04989 \text{ m}^3$, $V_2 = 2V_1 = 0.09977 \text{ m}^3$: [1]

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 600 \times \left(\frac{1}{2} \right)^{2/3}$$

$$\left(\frac{1}{2} \right)^{2/3} = \frac{1}{2^{2/3}} = \frac{1}{1.587} = 0.6300$$

$$T_2 = 600 \times 0.6300 = 378 \text{ K}$$

[1]

(ii) Using the ideal gas law for the final state: [1]

$$P_2 = \frac{nRT_2}{V_2} = \frac{1 \times 8.314 \times 378}{0.09977} = \frac{3143}{0.09977}$$

$$P_2 = 3.15 \times 10^4 \text{ Pa}$$

[1]

This is **lower** than the initial pressure of $1.0 \times 10^5 \text{ Pa}$ (by roughly a factor of 3). This makes physical sense: the gas expanded while also cooling.

Verification using $PV^\gamma = \text{const}$: $P_1 V_1^\gamma = 1.0 \times 10^5 \times (0.04989)^{5/3}$. $(0.04989)^{5/3} = (0.04989)^1 \times (0.04989)^{2/3} = 0.04989 \times 0.1357 = 0.006772$. So $P_1 V_1^\gamma = 677.2$. $P_2 V_2^\gamma = 3.15 \times 10^4 \times (0.09977)^{5/3} = 3.15 \times 10^4 \times 0.02151 = 677.6 \checkmark$

(d) (i) Carnot efficiency: (*From formula sheet*)

$$\eta = 1 - \frac{T_c}{T_h} = 1 - \frac{300}{500} = 1 - 0.6$$

$$\eta = 0.40 = 40\%$$

[1]

(ii) Work done per cycle: (*Recalled — not on formula sheet*)

$$W = \eta Q_h = 0.40 \times 1000$$

$$W = 400 \text{ J}$$

[1]

(iii) Heat rejected to cold reservoir: (*Recalled — not on formula sheet*)

$$Q_c = Q_h - W = 1000 - 400$$

$$Q_c = 600 \text{ J}$$

[1]

(iv) Entropy changes: (*Recalled — not on formula sheet*)

$$\Delta S_{\text{hot}} = -\frac{Q_h}{T_h} = -\frac{1000}{500} = -2 \text{ J/K (heat leaves hot reservoir)}$$

$$\Delta S_{\text{cold}} = +\frac{Q_c}{T_c} = +\frac{600}{300} = +2 \text{ J/K (heat enters cold reservoir)}$$

$$\Delta S_{\text{universe}} = \Delta S_{\text{hot}} + \Delta S_{\text{cold}} = -2 + 2 = 0 \text{ J/K}$$

$\Delta S_{\text{hot}} = -2 \text{ J/K}$, $\Delta S_{\text{cold}} = +2 \text{ J/K}$, $\Delta S_{\text{universe}} = 0$ (Carnot engine is reversible).

[1]

Question 3.

[15 marks]

Direct SHM and Wave Applications(a) (i) The angular frequency is: (*From formula sheet*)

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{100}{0.25}} = \sqrt{400} \quad [1]$$

$$\omega_0 = 20 \text{ rad/s}$$

The period is:

$$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{20}$$

$$T = 0.314 \text{ s}$$

[1]

(ii) The total energy of the oscillator is: (*Recalled — not on formula sheet*)

$$E = \frac{1}{2}kA^2 = \frac{1}{2} \times 100 \times (0.04)^2 = \frac{1}{2} \times 100 \times 0.0016$$

$$E = 0.080 \text{ J} = 80 \text{ mJ}$$

[1]

(iii) The maximum speed occurs at $x = 0$, where all energy is kinetic: (*Recalled — not on formula sheet*)

$$v_{\max} = A\omega_0 = 0.04 \times 20$$

$$v_{\max} = 0.80 \text{ m/s}$$

[1]

$$\text{Check: } \frac{1}{2}mv_{\max}^2 = \frac{1}{2}(0.25)(0.64) = 0.080 \text{ J} = E \quad \checkmark$$

(b) (i) Comparing $y(x, t) = 0.04 \sin(3x - 12t)$ with the standard form $y = A \sin(kx - \omega t)$ (*From formula sheet*):

- Amplitude: $A = 0.04 \text{ m}$ [1]
- Wavenumber: $k = 3 \text{ rad/m}$, so wavelength $\lambda = \frac{2\pi}{k} = \frac{2\pi}{3} = 2.09 \text{ m}$ (*Recalled — not on formula sheet*)
- Angular frequency: $\omega = 12 \text{ rad/s}$, so frequency $f = \frac{\omega}{2\pi} = \frac{12}{2\pi} = 1.91 \text{ Hz}$ (*Recalled — not on formula sheet*)
- Wave speed: $v = \frac{\omega}{k} = \frac{12}{3} = 4.0 \text{ m/s}$ (or $v = f\lambda = 1.91 \times 2.09 = 4.0 \text{ m/s}$)

$$A = 0.04 \text{ m}, \lambda = 2\pi/3 \approx 2.09 \text{ m}, f = 6/\pi \approx 1.91 \text{ Hz}, v = 4.0 \text{ m/s}.$$

[1]

(ii) The wave has the form $\sin(kx - \omega t)$, which corresponds to a wave travelling in the **positive x -direction**. [1](A wave of the form $\sin(kx + \omega t)$ would travel in the negative x -direction.)

- (c) (i) For a string fixed at both ends, the fundamental frequency is: (*From formula sheet*)

$$f_1 = \frac{v}{2L} = \frac{200}{2 \times 2} = \frac{200}{4}$$

$$f_1 = 50 \text{ Hz}$$

[1]

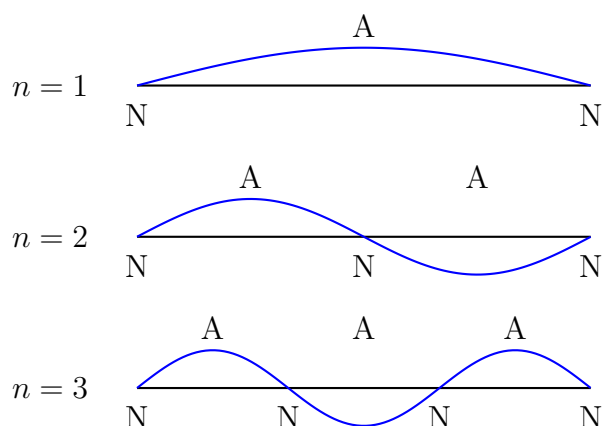
- (ii) The harmonics are $f_n = n f_1$ (*From formula sheet*):

$$f_2 = 2 \times 50 = 100 \text{ Hz}, \quad f_3 = 3 \times 50 = 150 \text{ Hz}$$

[1]

- (iii)

[2]



Key features:

- 1st harmonic: 2 nodes (both ends), 1 antinode (centre).
- 2nd harmonic: 3 nodes (both ends + centre), 2 antinodes.
- 3rd harmonic: 4 nodes (both ends + two interior), 3 antinodes.

- (d) (i) Source approaching: using $f' = f \frac{v}{v - v_s}$ (*From formula sheet*) (stationary observer, source approaching $\Rightarrow$ denominator uses minus sign):

[1]

$$f' = 800 \times \frac{340}{340 - 30} = 800 \times \frac{340}{310}$$

$$f' = 800 \times 1.0968 = 877 \text{ Hz}$$

[1]

- (ii) Source receding: using $f' = f \frac{v}{v + v_s}$ (*From formula sheet*) (stationary observer, source receding $\Rightarrow$ denominator uses plus sign):

[1]

$$f' = 800 \times \frac{340}{340 + 30} = 800 \times \frac{340}{370}$$

$$f' = 800 \times 0.9189 = 735 \text{ Hz}$$

[\[1\]](#)

Note: The approaching frequency (877 Hz) is higher than the emitted frequency, and the receding frequency (735 Hz) is lower, as expected.

Question 4.

[15 marks]

Basic Dirac Notation and Spin Measurements

- (a) (i) The normalisation condition requires $\langle\psi|\psi\rangle = 1$: (*Recalled — not on formula sheet*)

$$\langle\psi|\psi\rangle = \left|\frac{3}{5}\right|^2 + \left|\frac{4}{5}\right|^2 = \frac{9}{25} + \frac{16}{25} = \frac{25}{25} = 1 \quad \checkmark$$

The state is normalised since $|3/5|^2 + |4/5|^2 = 9/25 + 16/25 = 1$.

[1]

- (ii) The state $|\psi\rangle = \frac{3}{5}|\uparrow_z\rangle + \frac{4}{5}|\downarrow_z\rangle$ is already expressed in the S_z eigenbasis.

By the Born rule (*Recalled — not on formula sheet*):

- $S_z = +\hbar/2$ (spin up): probability = $|3/5|^2 = 9/25 = 0.36$
- $S_z = -\hbar/2$ (spin down): probability = $|4/5|^2 = 16/25 = 0.64$

$$P(S_z = +\hbar/2) = 9/25; \quad P(S_z = -\hbar/2) = 16/25.$$

[2]

- (b) From the formula sheet: (*From formula sheet*)

$$|\uparrow_x\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle), \quad |\downarrow_x\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle - |\downarrow_z\rangle)$$

Inverting these relations (add and subtract):

$$|\uparrow_z\rangle = \frac{1}{\sqrt{2}}(|\uparrow_x\rangle + |\downarrow_x\rangle), \quad |\downarrow_z\rangle = \frac{1}{\sqrt{2}}(|\uparrow_x\rangle - |\downarrow_x\rangle)$$

[1]

Substituting into $|\psi\rangle = \frac{3}{5}|\uparrow_z\rangle + \frac{4}{5}|\downarrow_z\rangle$:

$$\begin{aligned} |\psi\rangle &= \frac{3}{5} \cdot \frac{1}{\sqrt{2}}(|\uparrow_x\rangle + |\downarrow_x\rangle) + \frac{4}{5} \cdot \frac{1}{\sqrt{2}}(|\uparrow_x\rangle - |\downarrow_x\rangle) \\ &= \frac{1}{5\sqrt{2}} \left[(3+4)|\uparrow_x\rangle + (3-4)|\downarrow_x\rangle \right] \end{aligned}$$

$$|\psi\rangle = \frac{7}{5\sqrt{2}}|\uparrow_x\rangle - \frac{1}{5\sqrt{2}}|\downarrow_x\rangle$$

[1]

Probability of measuring $S_x = +\hbar/2$:

$$P(S_x = +\hbar/2) = \left| \frac{7}{5\sqrt{2}} \right|^2 = \frac{49}{50}$$

$$P(S_x = +\hbar/2) = 49/50 = 0.98$$

[1]

Check: $P(S_x = -\hbar/2) = |-1/(5\sqrt{2})|^2 = 1/50$. Sum = $49/50 + 1/50 = 1 \checkmark$

(c) (i) The operator $\hat{A} = |\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|$.

The matrix elements are: $A_{11} = \langle\uparrow|\hat{A}|\uparrow\rangle = 1$, $A_{12} = \langle\uparrow|\hat{A}|\downarrow\rangle = 0$, $A_{21} = \langle\downarrow|\hat{A}|\uparrow\rangle = 0$, $A_{22} = \langle\downarrow|\hat{A}|\downarrow\rangle = -1$.

$$\hat{A} \leftrightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z$$

This is the Pauli z -matrix.

[1]

(ii) Since the matrix is already diagonal, the eigenvalues and eigenvectors can be read off directly:

$$\text{Eigenvalue } +1 \text{ with eigenvector } |\uparrow_z\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; \quad \text{Eigenvalue } -1 \text{ with eigenvector } |\downarrow_z\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

[2]

(iii) $\hat{A}$ is Hermitian. The matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is real and symmetric, so its conjugate transpose equals itself:

$$\hat{A}^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^*{}^T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \hat{A}$$

Yes, $\hat{A}$ is Hermitian since $\hat{A}^\dagger = \hat{A}$ (equivalently, the matrix is real and symmetric).

[1]

(d) The state is $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$.

From the formula sheet: (*From formula sheet*)

$$|\uparrow_x\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$$

Therefore $|\psi\rangle = |\uparrow_x\rangle$.

$$|\psi\rangle = |\uparrow_x\rangle, \text{ which is an eigenstate of } S_x.$$

[1]

$$S_x |\uparrow_x\rangle = +\frac{\hbar}{2} |\uparrow_x\rangle$$

The eigenvalue is $+\hbar/2$.

[1]

For the expectation value of S_z on this state: (*Recalled — not on formula sheet*)

$$\langle S_z \rangle = \langle\psi| S_z |\psi\rangle$$

Since $|\psi\rangle = \frac{1}{\sqrt{2}} |\uparrow_z\rangle + \frac{1}{\sqrt{2}} |\downarrow_z\rangle$, and $S_z = \frac{\hbar}{2}(|\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|)$ (*From formula sheet*):

[1]

$$\langle S_z \rangle = \frac{\hbar}{2} \left(\left| \frac{1}{\sqrt{2}} \right|^2 - \left| \frac{1}{\sqrt{2}} \right|^2 \right) = \frac{\hbar}{2} \left(\frac{1}{2} - \frac{1}{2} \right)$$

$$\langle S_z \rangle = 0$$

[1]

This makes physical sense: the state has equal probability of being measured as spin-up or spin-down in the z -direction, so the average is zero.